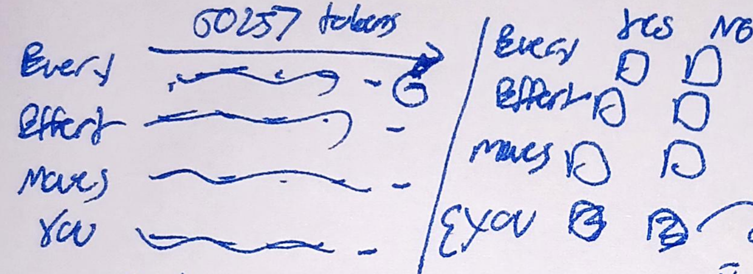
Model Initialization with Pretrained weights

○ Prepare model for classification finetuning

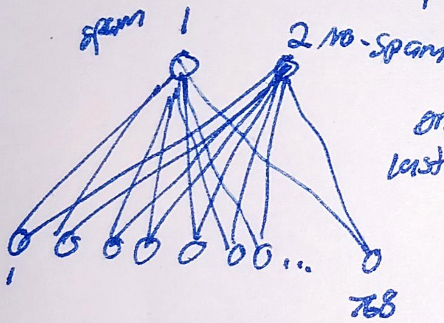
Stage 2: 4) Initialize Model → 5) Load pretrained weights from GPT-2 → 6) Modify model for finetuning

Step 1: Load pretrained weights

Step 2: Modify architecture by adding a classification head



→ Look at final row "you" in this case to get final answer. The last row contains all the previous information, we get two values corresponding to "yes" "no". These values are indicative of probabilities, so based on whichever is higher, we classify as "spam" or "no spam"



only finetune last transformer block

we replace the original linear output layer above w/ a layer that maps from 768 hidden units to only 2 units, where the 2 units represent 2 classes (spam and non-spam)

↳ classification Head

Step 3: Select which layers you want to finetune

↳ Since we start w/ a pretrained model, it isn't necessary to finetune all layers.

↳ This is because the lower layers capture basic language structures and semantics, which are applicable across a wide range of tasks and datasets.

↳ finetuning last layers is enough

NEXT PAGE →

We will finetune:
- final output head
- final transformer block
- final LayerNorm module

} We will freeze all other parameters

Why do we always extract the last output token?

	Do	you	have	time
Do	1.0			
you	.55	.45		
have	.38	.30	.32	
time	.27	.24	.24	.25

Tokens masked out via causal attention mask

Last token is the only token with an attention score to all other tokens

	Yes	No
Do		
you		
have		
time		